Three-Dimensional Icing Code Development Status

presented
by

Colin Bidwell

NASA Glenn Research Center
Icing Branch
Cleveland, Ohio

The following presentation will summarize the status of the Three-Dimensional Icing Code Development at Glenn Research Center. A quick overview of the methodology used in the code will be followed by a summary of past efforts. Some preliminary results from the fully 3 Dimensional ice accretion code ICE3D will be presented.

Overview

- LEWICE3D
- Post Processors
- Demonstration
- Future Plans
- Conclusions

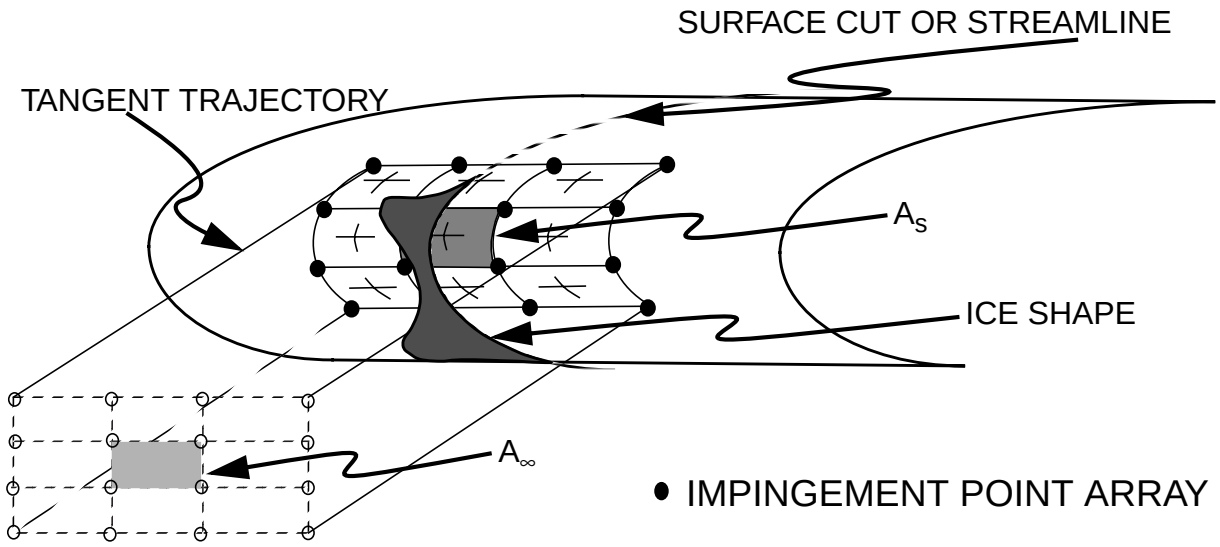
In the following a description of the LEWICE3D grid based ice accretion code will be given and some results will be presented for the different LEWICE3D interfaces. A discussion of the LEWICE3D post processors, PLTLEW3D, LEWPLT3D and GRPHCNV will be followed by a demonstration of LEWICE3D program for a Boeing 737-300 inlet. The ICE3D ice accretion program will be discussed under the Future Plans section. Finally some general conclusions about the work to date will be given.

LEWICE3D

- User supplies grid based flow solution. LEWICE3D can handle multi-block structured grids, “VSAERO” type structured grids, adaptive cartesian grids (ICEGRID), and unstructured grids.
- Trajectories are calculated using Adams-type predictor-corrector method developed by Hillyer Norment.
- Tangent trajectories and collection efficiencies for simple 2D or 3D regions can be calculated using a modified LEWICE2D scheme.
- Tangent trajectories and collection efficiencies for complex regions are calculated using a new unstructured Monte-Carlo based collection efficiency method.
- Streamlines are calculated using a 4th order Runge-Kutta integration scheme.
- Ice growth calculated using modified LEWICE2D scheme.

LEWICED3D is a grid based 3-Dimensional ice accretion code which has been under development for 9 years. Unlike the LEWICE2D code the LEWICE3D program does not come with a flow solver. The program can interface with just about any type of flow solution including: structured and unstructured grid based solutions, and panel code based solutions using the ICEGRID interface. The code combines several algorithms to generate ice shapes using a strip approach. The program can calculate trajectories, streamlines, and ice shapes for a variety of flows including: wing, inlets, ducts, radomes and bodies of rotation such as fuselage noses.

LEWICE3D Methodology



- ● IMPINGEMENT POINT ARRAY
- ○ RELEASE POINTS FOR IMPINGING TRAJECTORIES
- \+ ARRAY OF CENTROIDS OF IMPACT AREAS ON SURFACE

$\beta = A_\infty/A_S$

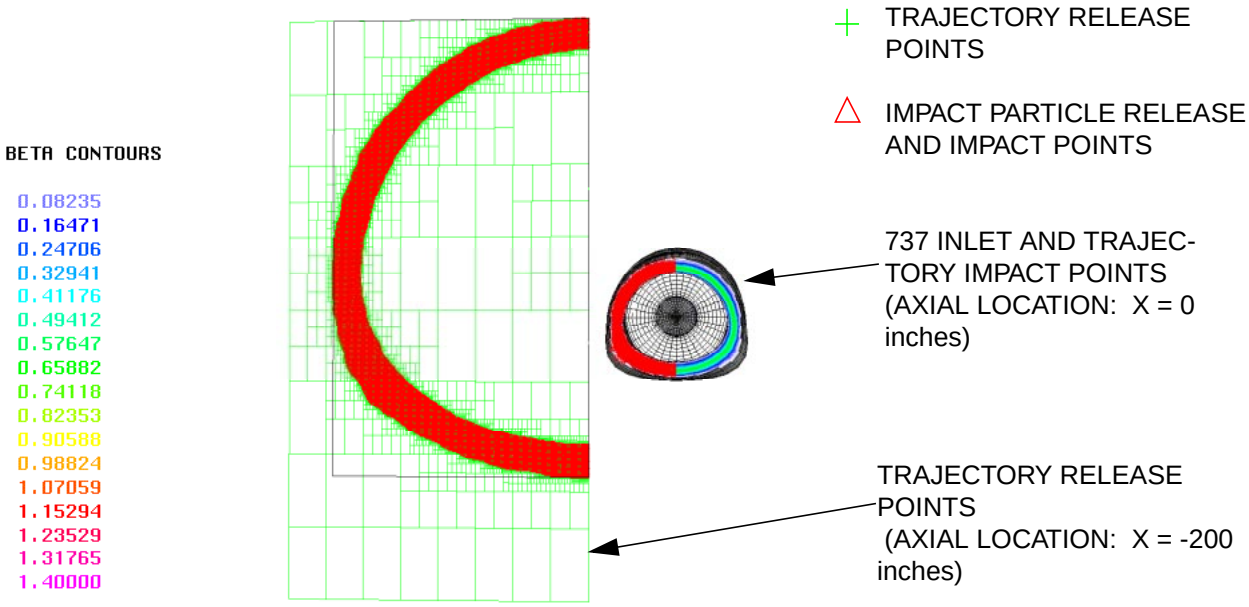
This viewgraph shows a graphic of the LEWICE3D jobstream. Shown is a region of interest which could be a swept leading edge or an inlet lip. The first step in the calculation involves generating the surface streamline or surface cut at the section-of-interest. The next step involves an iteration to determine the tangent trajectories. After this a matrix of particles is released to determine the surface collection efficiencies. A surface interpolation is then carried out to determine the collection efficincies and edge velocities along the surface cut. Finally a modified LEWICE2D ice accretion is carried out along the surface streamline to determine the ice thickness distribution and the resulting ice shape.

LEWICE3D Unstructured Collection Efficiency Calculation

- Method combines tangent trajectory search and impingement efficiency calculation into one step using modified “Quad-Tree” searching scheme and Monte-Carlo based collection efficiencies.
- Method has improved speed over “scan-line” Monte-Carlo collection efficiency methods (5-100 times faster).
- Method can handle highly complex surfaces with a minimal of user input.
- Method is amenable to parallel processing.
- Generates whole body collection efficiencies which are required by any full 3D ice accretion code.
- Algorithm picks up impact regions that traditional methods may miss if not expected.

The unstructured collection efficiency method was developed to allow the calculation of overall collection efficiencies for complex bodies with a minumum of user input and with a minimum of computational resources. The user can develop whole body collection efficiencies by simply specifying a bounding box. The user does not have to worry about setting up impingment limit searches for complex inpingement regions (e.g. ducts) or worry about missing a region of impact because it was not anticipated (e.g. radome or wing flaps). The method has an iteration algorithm which targets trajectories which are near a body or which hit the body resulting in a minimum of calculated trajectories and hence execution time.

Unstructured Collection Efficiency Calculations For A Boeing 737-300 Inlet



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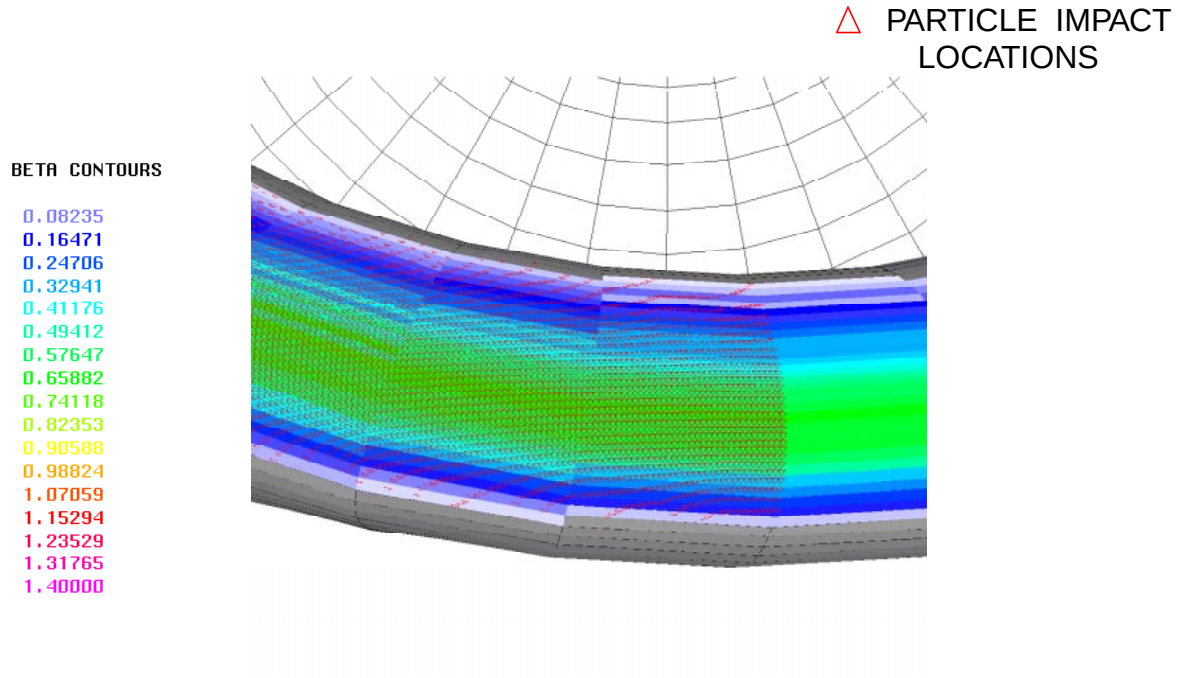
This figure illustrates the unstructured collection efficiency calculation for a Boeing 737-300 inlet. The collection efficiencies are calculated for each panel on the surface in a single pass. The algorithm releases trajectories within a bounding box specified by the user and recursively refines regions where trajectories hit the body or where they pass very close to the body until a minimum cell size is reached. The method uses the Monte-Carlo based method to calculate collection efficiency for each panel. That is, the collection efficiency for a surface panel is equal to the number of particles that hit the panel multiplied by the area per particle at the release plane divided by the area of the panel. The method combines the tangent trajectory calculation and the collection efficiency calculation into one step making it easier for users to calculate collection efficincies for complex surfaces. This case took about an hour to on a 400 Mhz Pentium II based processor and involved the calculation of approximately 13,000 trajectories. The method is slower than traditional methods because it requires a greater number of trajectories (100 -1000 times as many), but this is offset by the simplicity of the method and by the fact the calculation times are not prohibitive (several days on a fast PC for large cases).

Unstructured Collection Efficiency Calculations For A Boeing 737-300 Inlet



This figure shows a close up of the lip and is illustrative of the number of particles required for the unstructured calculation. The method requries that the area per particle is less than about one 10th of the panel area for convergence. This requirement results in about 20-50 particles per panel in regions of high collection efficiency.

LEWICE3D Panel Code Interface

- This interface allows LEWICE3D to be used with any first order panel method.
- User supplies surface model with velocities and grid with offbody velocities. The grid can be either a simple cartesian grid or an unstructured, cartesian, “Oct-tree” grid (ICEGRID).
- Typically several filters are needed to format the data for input to ICEGRID and LEWICE3D.
- Currently interfaces for VSAERO, PMARC, CMARC, QUADPAN, and Boeing A502 panel codes have been developed.

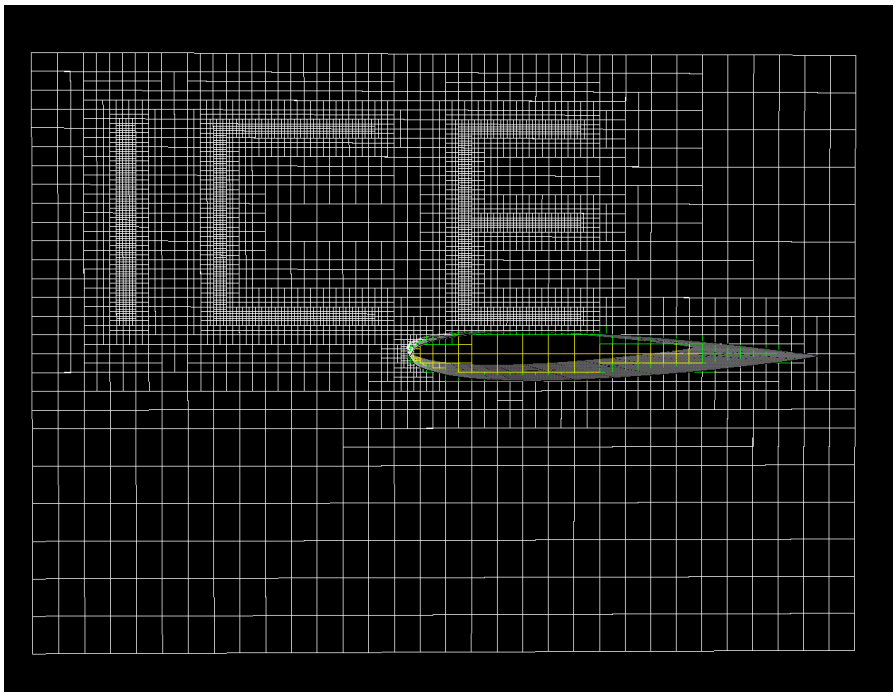
The LEWICE3D panel code interface requires the generation of a 3D grid with velocities. The grids can be either simple cartesian or of the unstructured cartesian type generated by ICEGRID. The ICEGRID code was developed at Glenn to enable the user to generate “good” trajectory grids about panel codes with a mimimum of effort. Several filters are necessary for the interface. These filters format the surface data and grid data into formats usable by a LEWICE3D and the users panel code. Filters have been developed for most of the panel codes in use today.

ICEGRID

- ICEGRID was developed at Glenn Research Center by Bidwell and Coirier to generate optimal grids for the LEWICE3D panel code interface.
- ICEGRID generates “good trajectory” grids with a minimal amount of input and with a minimum of grid points reducing the work load on the user and the users computer. The user inputs a file describing the surface geometry and a file describing the grid volume and refinement parameters.
- The code is similar to Oct-Tree methods in that it recursively divides the original grid volume until the refinement criteria for each cell has been met. The code can refine the grid near the geometry, near parts of the geometry, and near points or lines specified by the user.
- The code is different from Oct-Tree methods in that the grid volume can be multi-block, multiple-skewed and can have different refinement criteria in each direction.

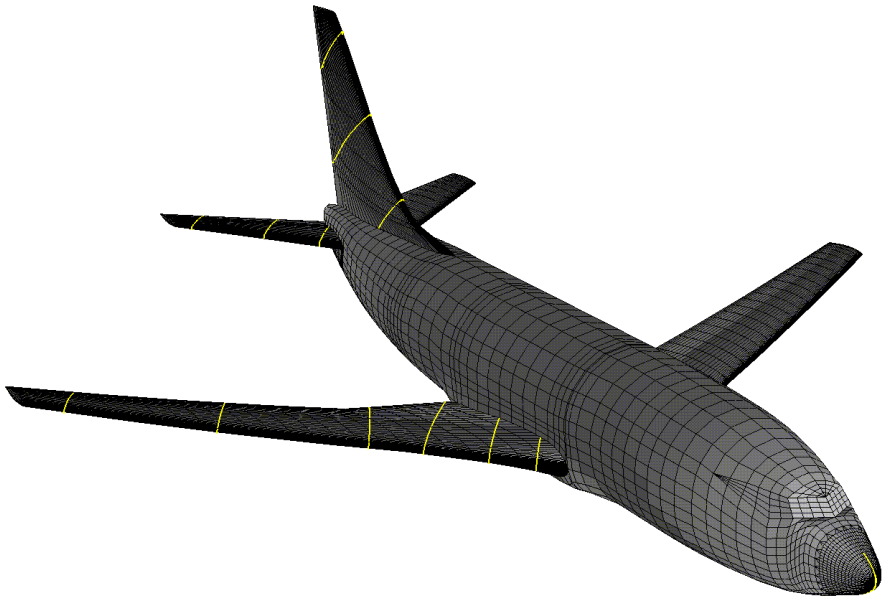
ICEGRID was developed for the panel code interface to LEWICE3D. ICEGRID is an adaptive 3D cartesian unstructured grid code. The program recursively refines a volume input by the user around objects of interest contained in the volume. The code can refine around lines and points and around parts of the surface model. The program will not sub-divide regions inside the body a feature which greatly reduces the size of the grids. The user inputs the minimum cell size near a surface and the maximum cell size far away from the surface. The code is similar to Oct-tree methods in that it recursively refines a region until cell size criterion are met. It is different from Oct-tree methods in that the code allows independent cell size criterion to be used in each of the three coordinate directions. This feature allows a signicant reduction in the size of the grids for cases such as swept wings where the user may want large span-wise cell sizes and small surface normal and chordwise cell sizes.

ICEGRID Example Grid Model



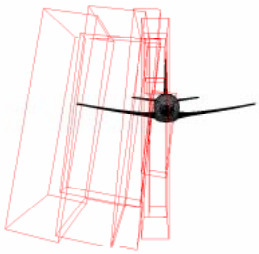
This shows an example of an ICEGRID generate grid about an unswept NACA-0012 wing. Featured are the ability to refine around lines and the leading edge of a surface. Also from the figure it can be seen that the code does not subdivide regions internal to the surface which results in smaller grids.

CMARC Panel Model With Sections-of-Interest

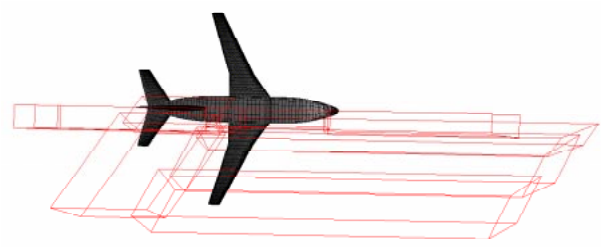


This figures depicts the CMARC panel model for the commercial transport along with the cuts where the icing analysis were done. A half model with 6545 panels was used for the analysis. Icing cuts were done on the wing, the vertical and horizontal tail, and on the nose of the aircraft.

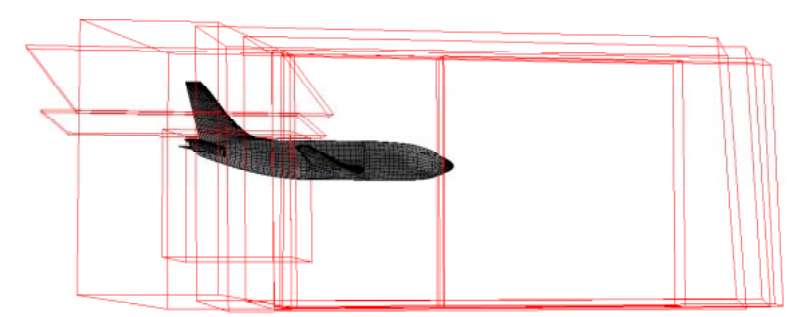
ICEGRID Grid Block Structure For A Commercial Transport



FRONT



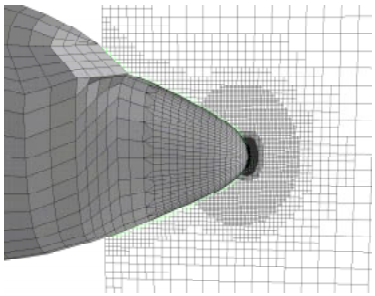
BOTTOM



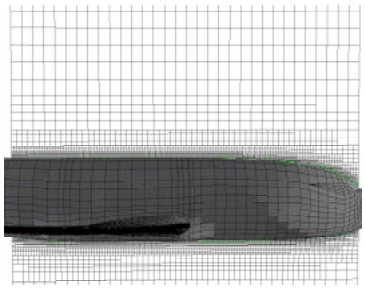
SIDE

The figure shows the ICEGRID blocking structure for the transport calculation. Nine blocks containing over 3 million points were used. The mult-block feature of the ICEGRID code was used to mimize the total number of points, the regrid time, and the memory and execution time required. The grids took aproximately 45 hours to generate. Each of the grids were calculated seperately and combined later into a single multi-block grid. This allowed the grids to be generated in parrallel on several different machines reducing the turn around time significantly. Each of the grid blocks was multiply skewed to reduce the size of the resulting grid. For wings with large changes in sweep, dihedral, and taper multiple grids with different skew angles must be used. In the above example 3 blocks were used to cover the main wing. The horizontal tail required only one block. From the figure it can be seen that the blocks are overlapping and are of various sizes. The grids are not used to calculate flows hence not much care need be given in the amount and location of the overlap. It is only important to insure that there are no gaps in the resulting blocking structure.

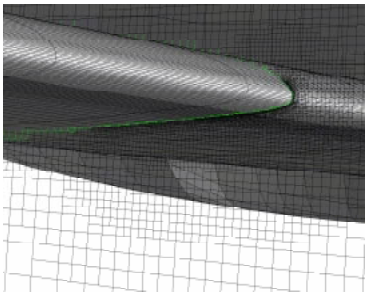
ICEGRID Grid Structure For A Commercial Transport



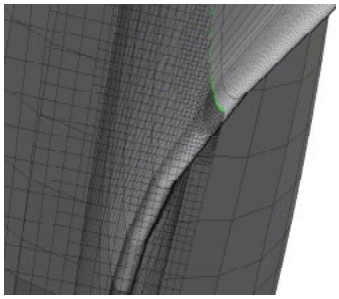
NOSE



FUSELAGE



WING

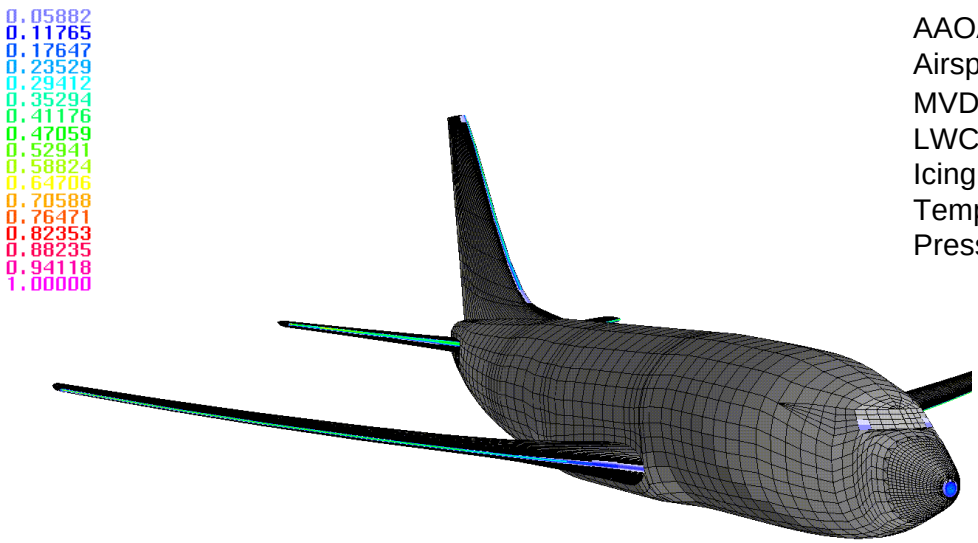


VERTICAL TAIL

This figure shows the grid structure used around various elements. For the nose and wing structures the points are clustered around the leading edges. For the wing elements small surface normal and chordwise spacing was used with larger spacing in the span-wise direction. For the nose element the same cell size was used in all three coordinate directions.The grid and panel spacing should be consistent. It is of no use to highly resolve a poor panel model and vice-versa. A “good” surface model for trajectory analysis typically requires 15-20 panels within the impingement region.

Overall Collection Efficiency For Commercial Transport

BETA CONTOURS



ICING CONDITIONS:

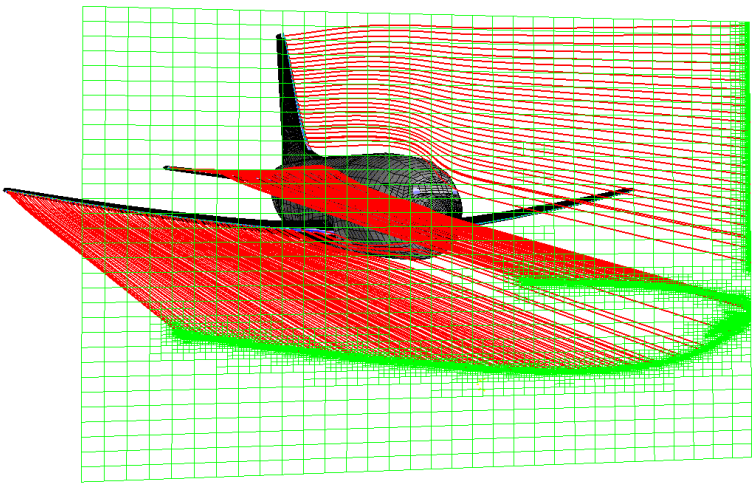
AAOA; 4 degrees
Airspeed; 135 m/s
MVD; 20 µm
LWC; .51 g/m^3
Icing time; 30 min.
Temperature; -14. C
Pressure; 57800 Pa

This figure shows the collection efficiency for the entire aircraft generated using the unstructured collection efficiency method. The calculation required about 20 hours of CPU time on a 400 Mhz, Pentium II based PC and involved the calculation of more than 300,000 trajectories. The calculation was relatively more expensive than the strip approach which took about 2 minutes but the method gives a more global picture of the collection efficiency of the aircraft. Areas of small or no impingement can be seen near the wing and tail root due to shadowing by the nose of the aircraft. The drop off in the maximum collection efficiency can be seen for the wing and tail as the root is approached due to the increase in chord.

Trajectories and Upstream Release Grid for Unstructured Collection Efficiency Method

BETA CONTOURS

0.05882
0.11765
0.17647
0.23529
0.29412
0.35294
0.41176
0.47059
0.52941
0.58824
0.64706
0.70588
0.76471
0.82353
0.88235
0.94118
1.00000

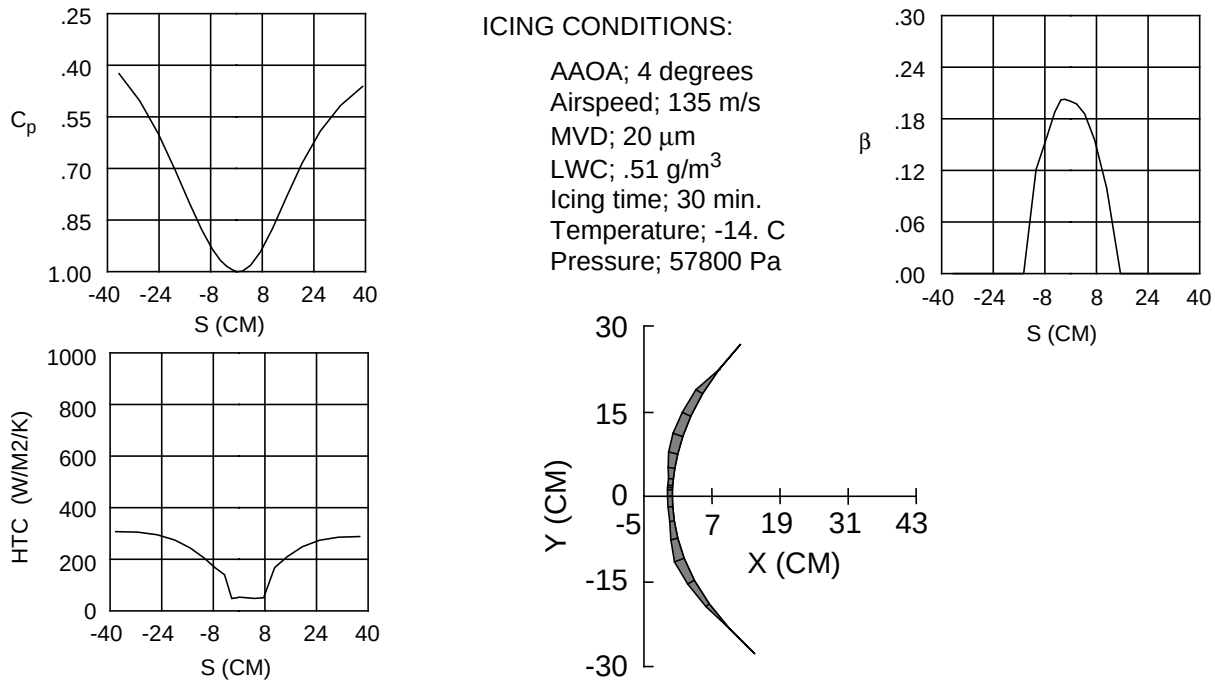


ICING CONDITIONS:

AAOA; 4 degrees
Airspeed; 135 m/s
MVD; 20 µm
LWC; .51 g/m^3
Icing time; 30 min.
Temperature; -14. C
Pressure; 57800 Pa

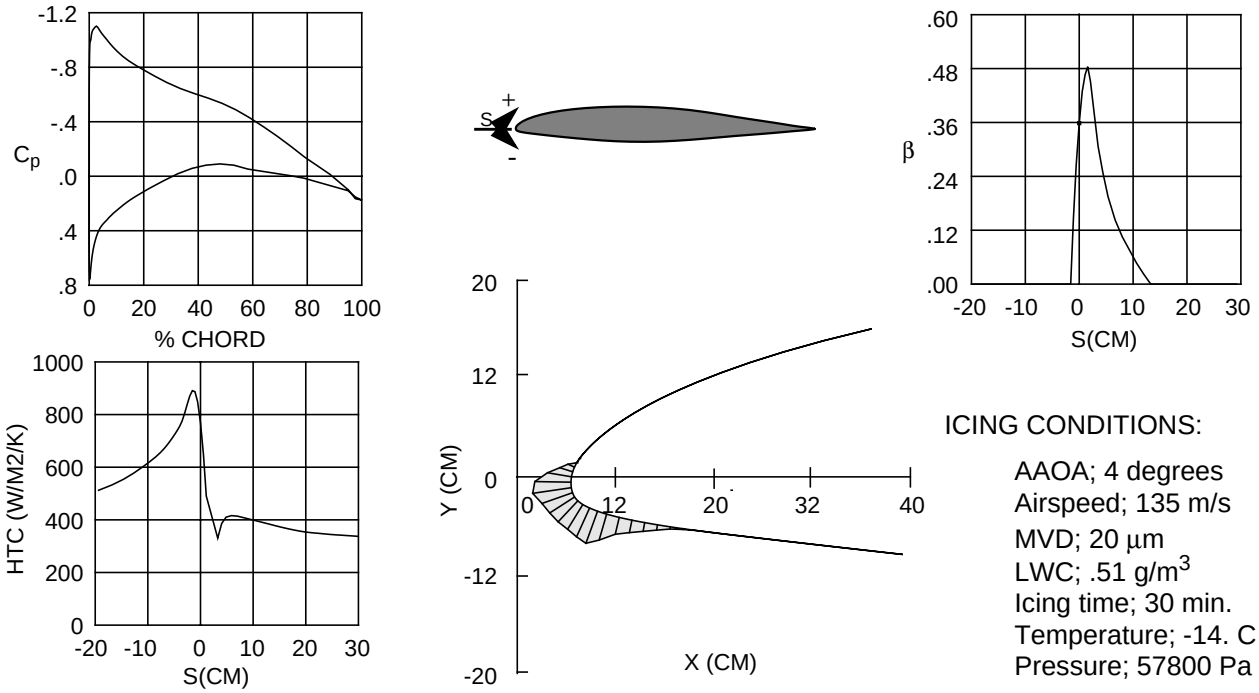
The figure shows some impact trajectories for the aircraft along with the trajectory release grid for the unstructured collection efficiency method. The source of the impacting trajectories and “shadowing” of the vertical tail root can be observed.

Ice Accretion Calculation for the Fuselage Nose



This figure depicts the icing calclulation for the nose section. Because of the blunt nature of the nose the collection efficiency and the heat transfer about the nose are relatively low. This results in a thin ice thickness distribution and a maximum thickness point which is relatively far back.

Ice Accretion Calculation For A Commercial Transport Wing
35% SPAN

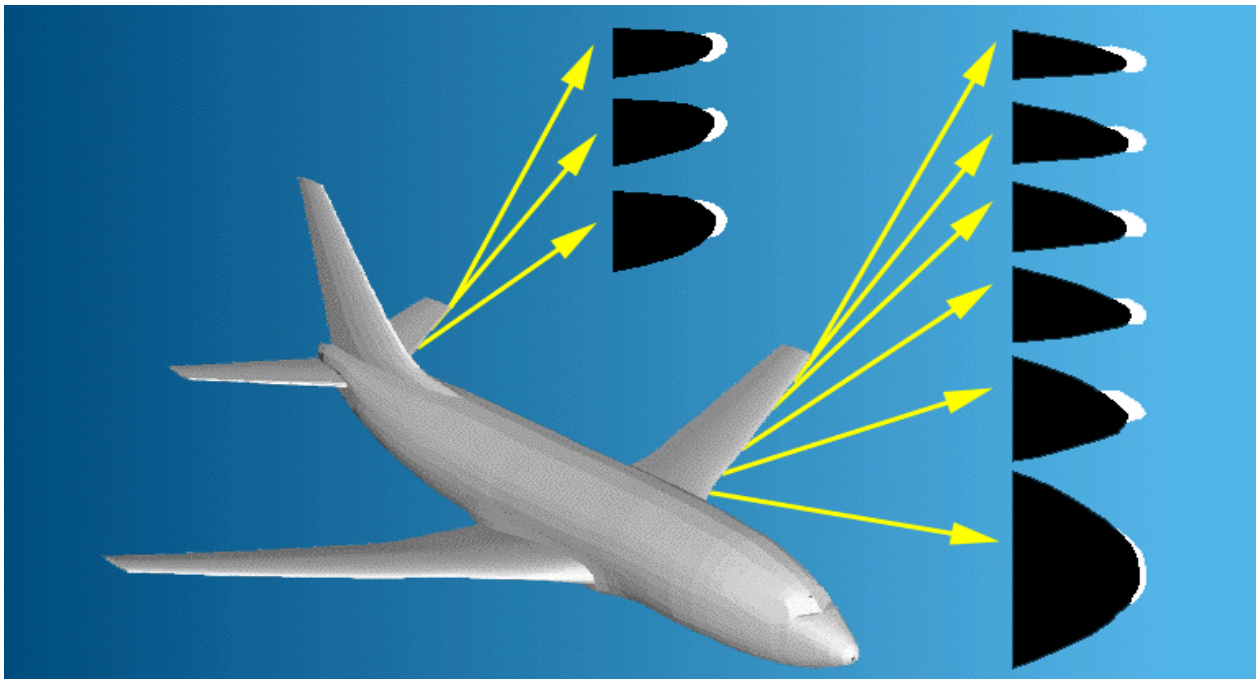


ICING CONDITIONS:

AAOA; 4 degrees
Airspeed; 135 m/s
MVD; 20 µm
LWC; .51 g/m^3
Icing time; 30 min.
Temperature; -14. C
Pressure; 57800 Pa

The icing calculation for the 35% station on the wing is shown. The case exhibits the features of a “mixed’ icing case having some run-back but no obvious horns.The heat transfer and collection efficiency at the leading edge are significantly higher for the wing than for the fuselage nose shown in the previous slide due to it’s sharper leading edge.

Ice Accretion Calculations For A Commercial Transport Airplane



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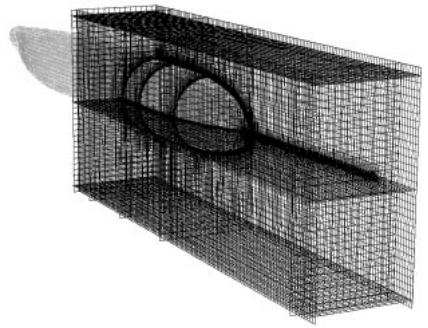
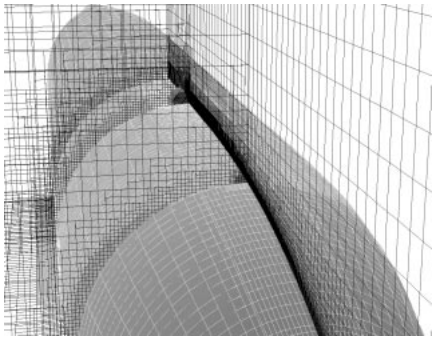
5-18

This figure depicts the LEWICE3D calculated ice shapes at several locations around the aircraft including the wing and the horizontal tail. The reduction in the maximum thickness of the ice with increased chord can be seen in the wing and tail ice shapes.

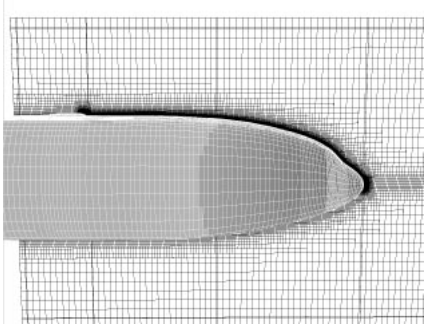
DC-10 Panel And ICEGRID Grid Models



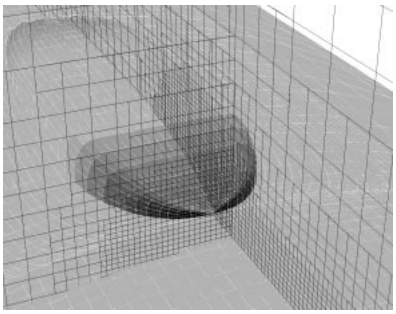
ISOMETRIC (FAR)



ISOMETRIC (CLOSE)



SIDE

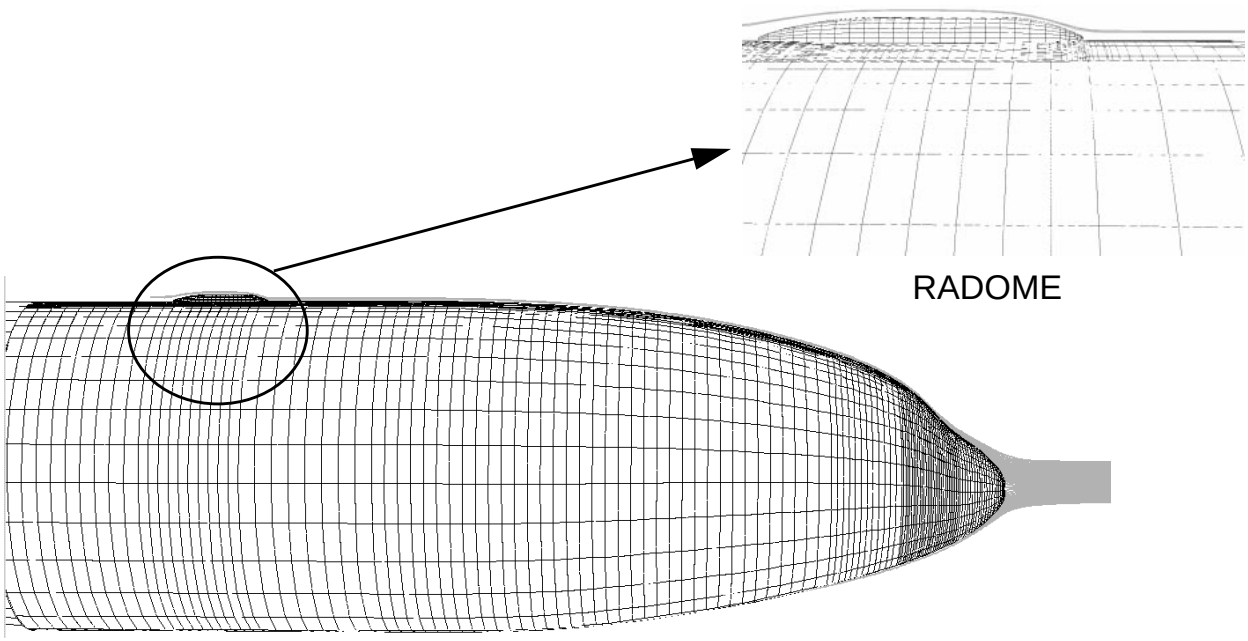


RADOME

The above shows the panel model and grid structure for a radome calculation. The clustering of cells and panels near the front of the radome and along the trajectory path leading up to the radome can be seen from the figure.

DC-10 Droplet Trajectory Calculations

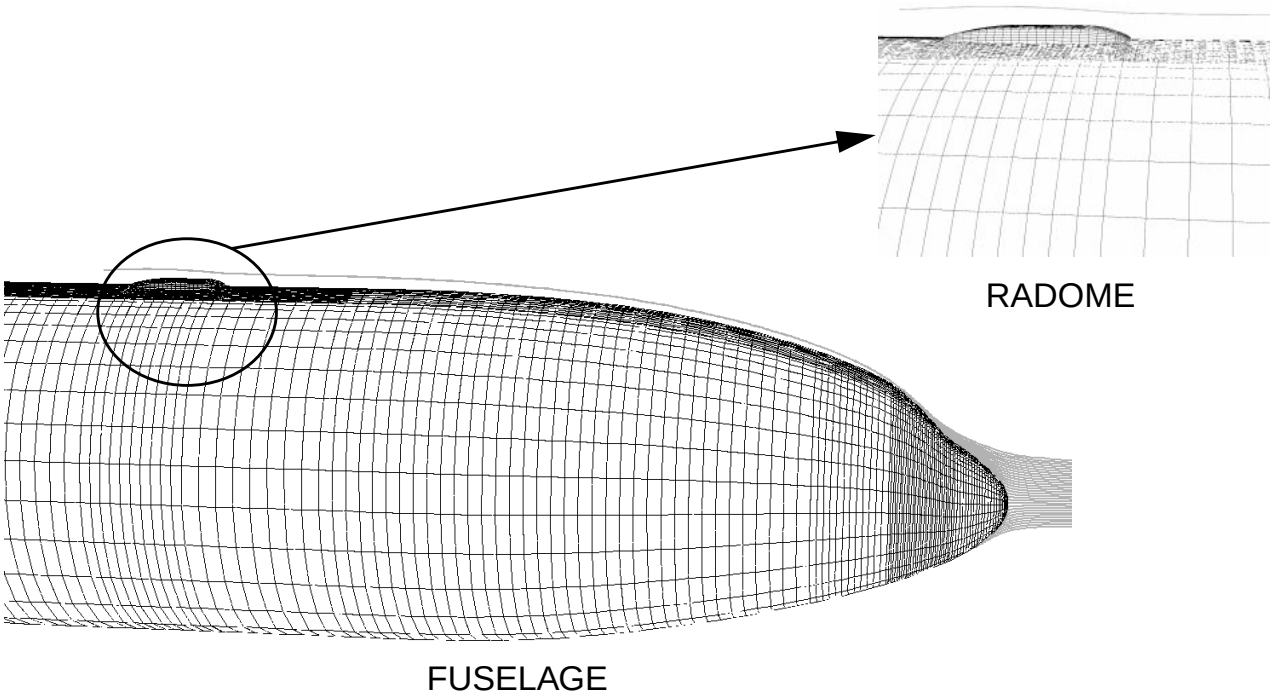
MVD 15 µm; Airspeed, 275 knots; AAOA, 0 degrees; Altitude, 15000 ft.; Temp, 28 ^{o}F



The above shows a series of trajectories around the radome for a 15 µm droplet. From the figure we can see that the particles come close but do not impact the radome. This is due to the relatively small inertia of the particles.

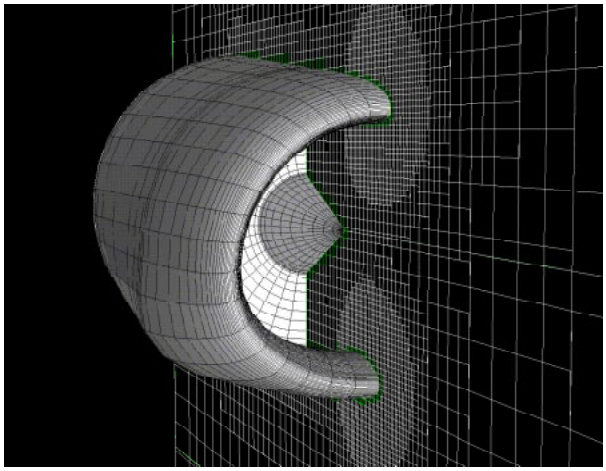
DC-10 Droplet Trajectory Calculations

MVD 40 µm; Airspeed, 275 knots; AAOA, 0 degrees; Altitude, 15000 ft.; Temp, 28 ^{o}F

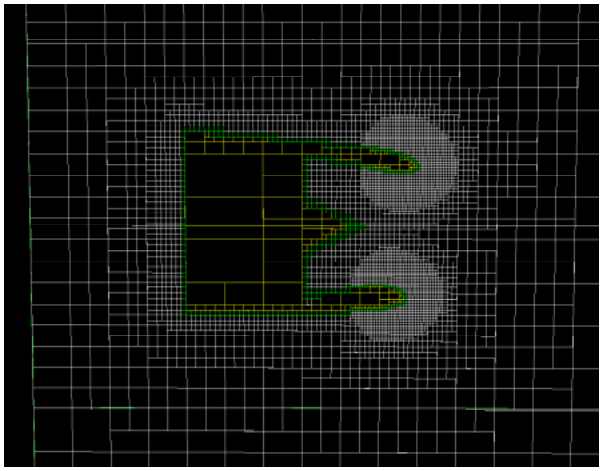


The above shows a series of trajectories around the radome for a 40 µm droplet. From the figure we can see that the particles either hit at the nose of the aircraft or are thrown out at the nose and and do not impact the radome. This is due to the relatively large inertia of the particles. The conclusion from this and the previous figure is that the radome will not collect any ice because the radome is conformal enough so that the smaller particles go around it and the large particles shadow it.

Boeing 737-300 Inlet Panel and ICEGRID Grid Models



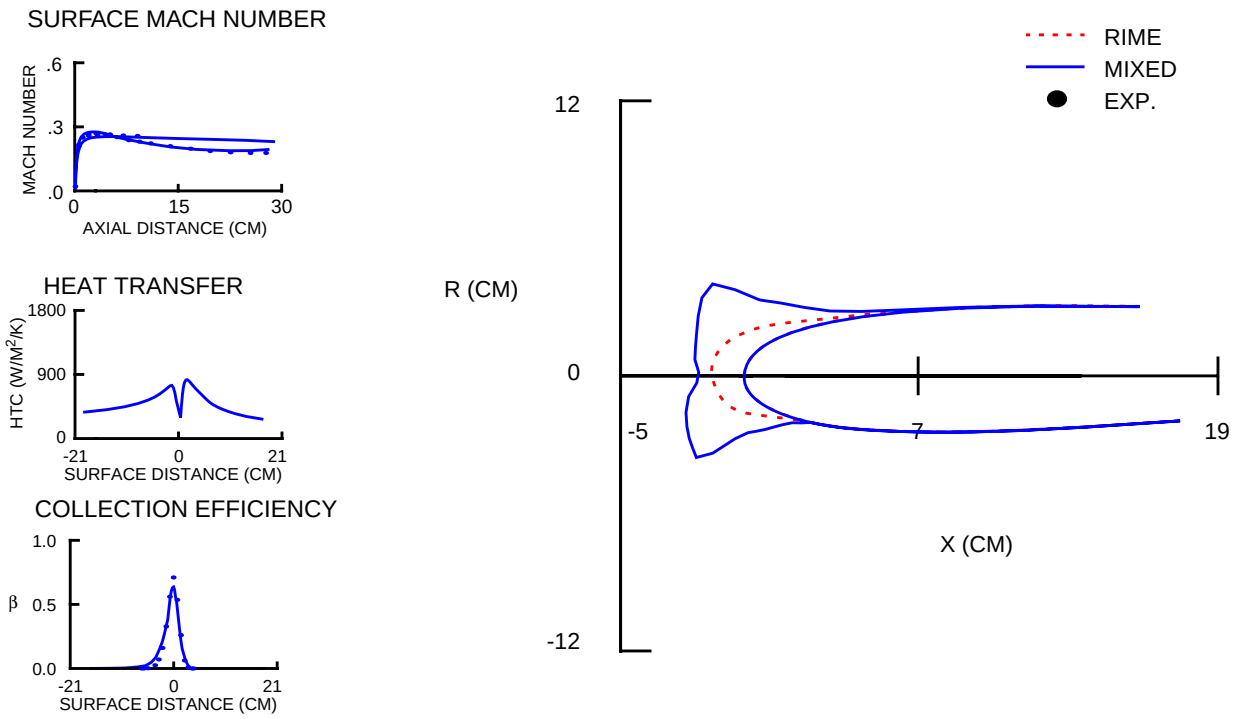
VSAERO PANEL MODEL AND GRID



GRID AT Y=0 PLANE

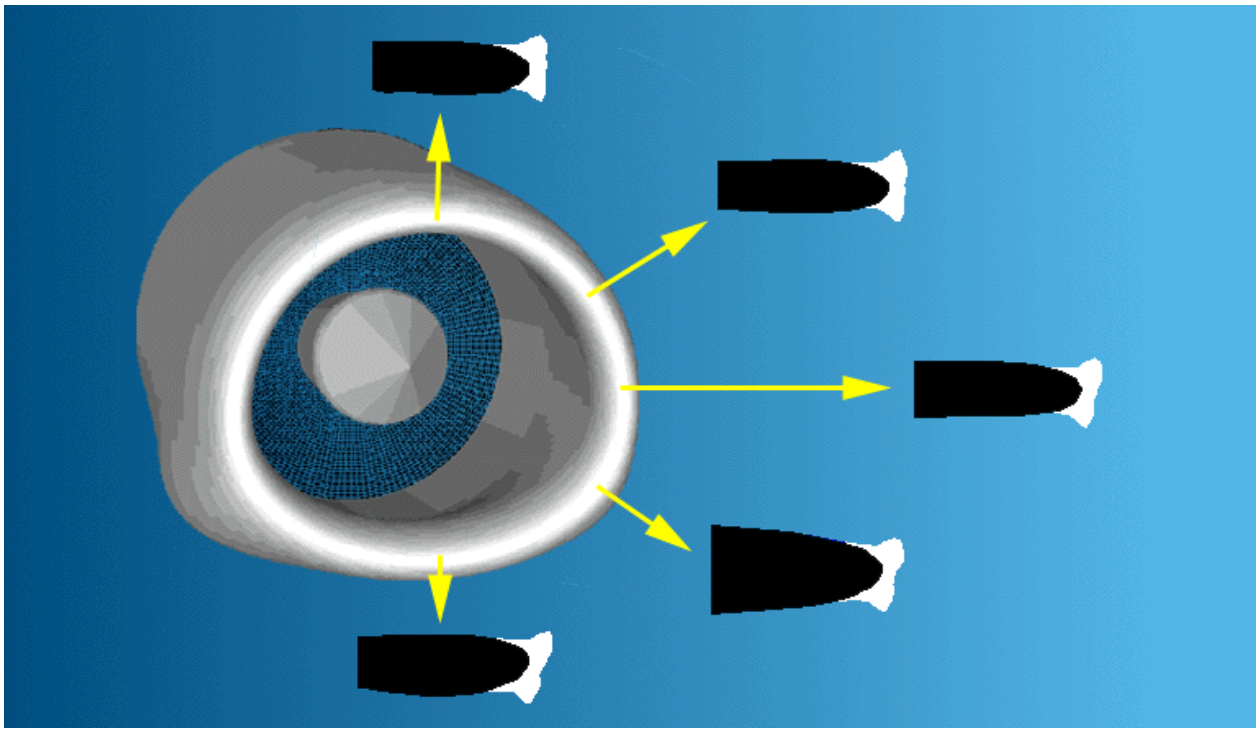
This figure shows the VSAERO panel model and ICEGRID grid model used for the Boeing 737-300 inlet. The clustering of panels and cells around the inlet lip can be seen in the above images. Also noteworthy is the absence of refinement internal to the surface of the inlet which results in a large reduction in the grid size.

Ice Accretion Calculations For A Boeing 737-300 Inlet



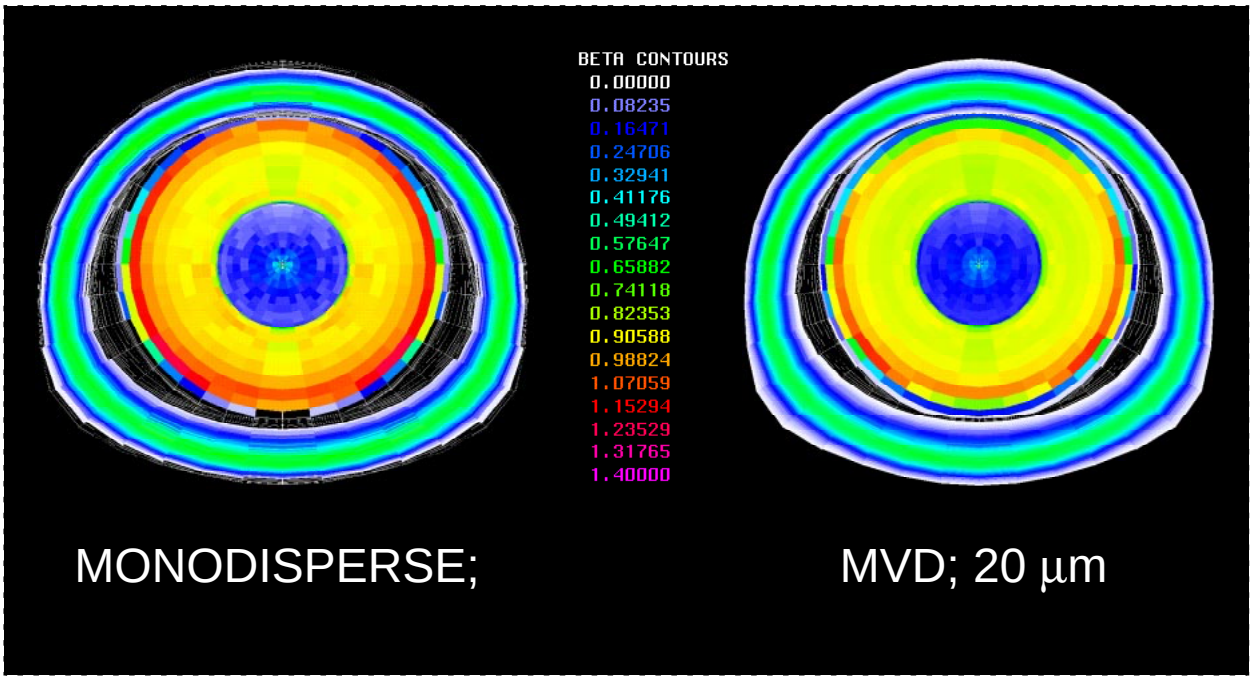
The above shows a rime and a mixed ice accretion calculation for the Boeing 737-300 inlet lip. Included are some comparisons to experimental surface Mach numbers and collection efficiencies. The comparisons to experiment are good. The calculated ice shapes are indicitive of a rime and a mixed ice shape. The rime shape is more conformal due to the absence of run-back. The mixed shape has some run-back but no significant horns.

Ice Accretion Calculations For A Boeing 737-300 Inlet



The above show the a series of calculated ice shapres around the lip of the Boeing 737-300 inlet. The variation in the shape around the inlet is due to the variation in the lip geometry and local angle-of-attack.

Collection Efficiency Calculations For A Boeing 737-300 Inlet (Effect Of Distribution)



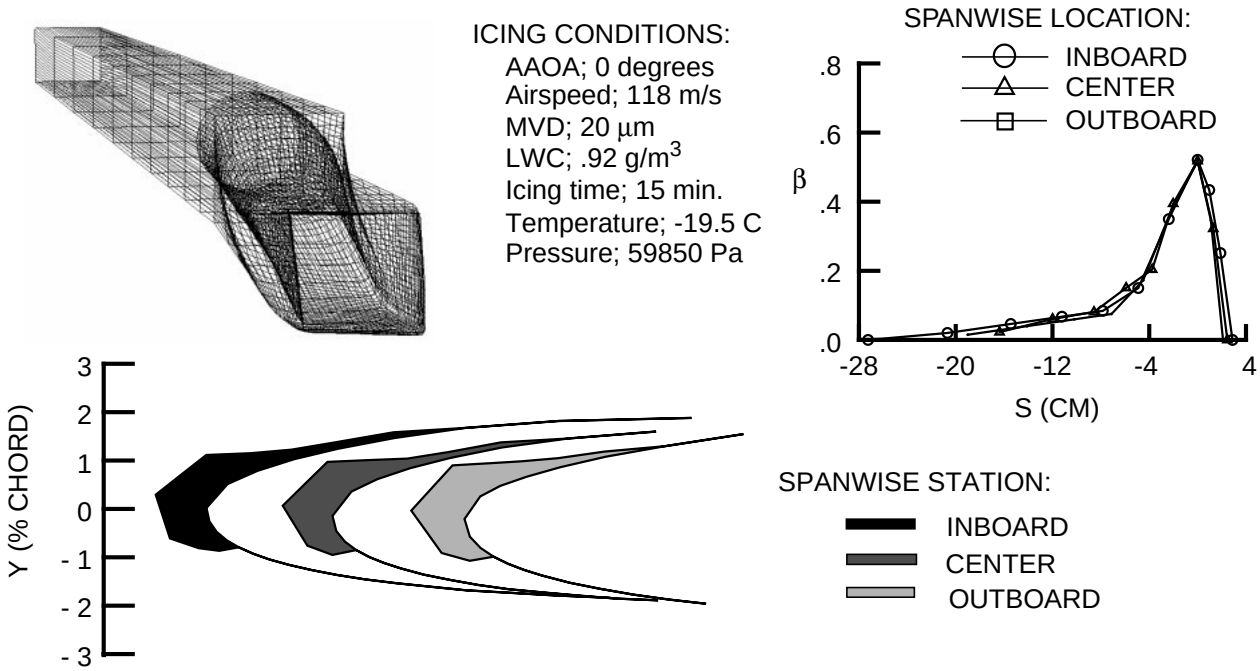
The above shows the overall collection efficiency for a Boeing 737-300 inlet for a single 20 µm droplet and for a 7 bin distribution with a 20 µm MVD. The effect of distrubution is seen more heavily at the compressor face than at the inlet lip. The collection efficiency distributions for the inlet lip are very similar. The monodisperse case has more shadowing at the compressor face than the distribution case. This is due to the absence of the smaller particles for the monodisperse case.

LEWICE3D Structured, Body Fitted Grid Interface

- This interface allows LEWICE3D to be used with any structured, multi-block, body-fitted Euler or Navier-Stokes flow solver.
- User inputs grid and flow solutions via PLOT3D formatted files. “I-Blanking” is used to describe solid boundaries and Chimera-type grids.
- Calculations have been made using NPARC, INS2D, OVERFLOW, ADPAC, WIND, and CFD-FASTRAN flow solutions.

The LEWICE3D program was originally developed to handle 3D structured, body fitted grids. The code has been used with many different flow solvers and can handle multi-block, Chimera grids, “iblanking” with Navier-Stokes or Euler solutions. The user supplies the grid and solution file in PLOT3D format.

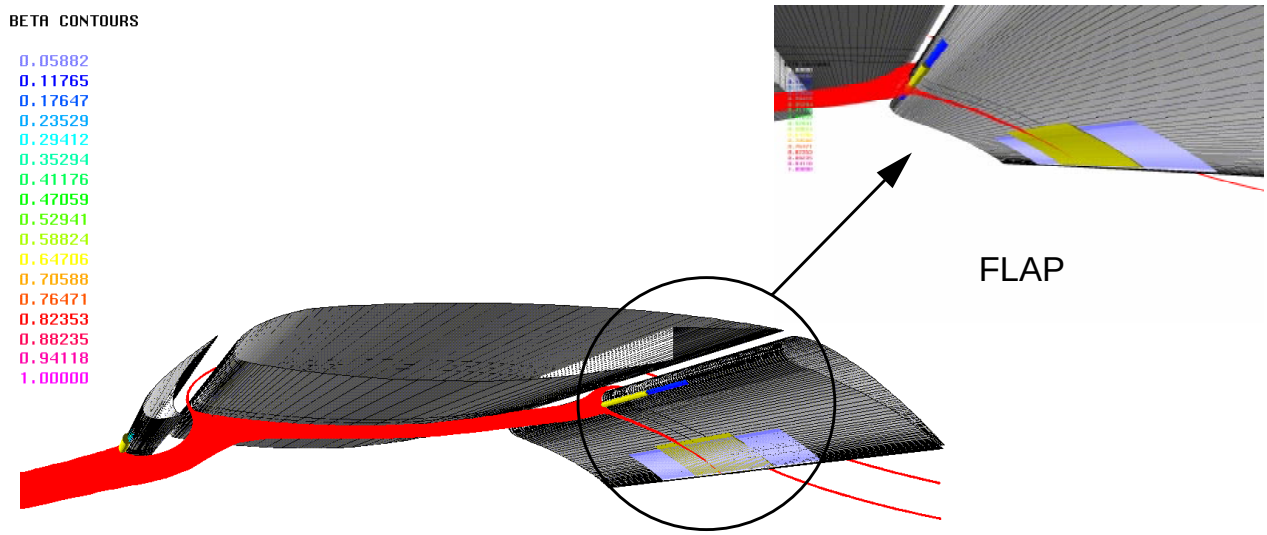
LEWICE3D Ice Accretion Calculations For F18 E/F Lower Inlet Lip At Several Spanwise Locations



This figure shows the grid and ice accretion calculations on the upper lip of the F18 E/F inlet. The Euler based solution represents one of the first calculations for the grid based LEWICE3D code. The collection efficiency and ice shape figures show very little span-wise variation along the upper lip.

Ice Accretion Calculations For A Multi-Element Wing

Landing Configuration, α; 0 degrees, Re; 17 million, Mach; .2
LWC; .7 g/m^3, MVD; 20 μm, Icing time; 45 minutes, K_{mod}; .00861

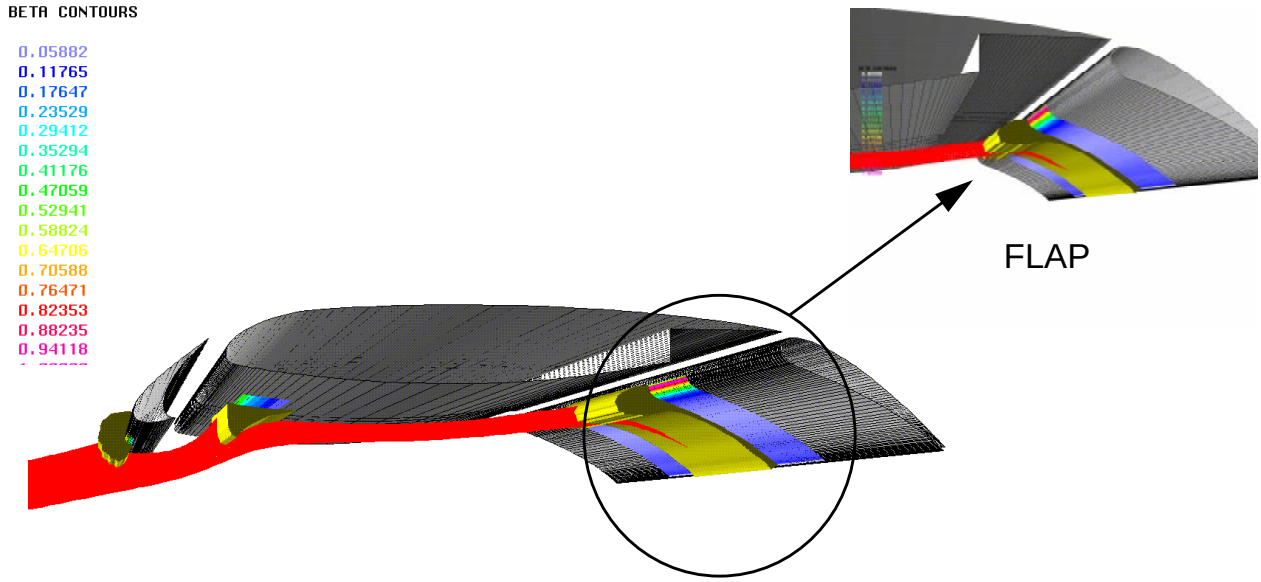


The above depicts an ice accretion calculation for a multi-element wing at full scale Reynolds Number under a fairly severe icing condition. The INS2D flow solver was used to generate the incompressible Navier-Stokes flow solution on the four block Chimera grid. Because of the size of the wing the severe icing condition actually results in very little ice buildup.

Ice Accretion Calculations For A Multi-Element Wing

Landing Configuration, α; 0 degrees, Re; 5 million, Mach; .2

LWC; .7 g/m^3, MVD; 20 µm, Icing time; 45 minutes, K_{mod}; .03820



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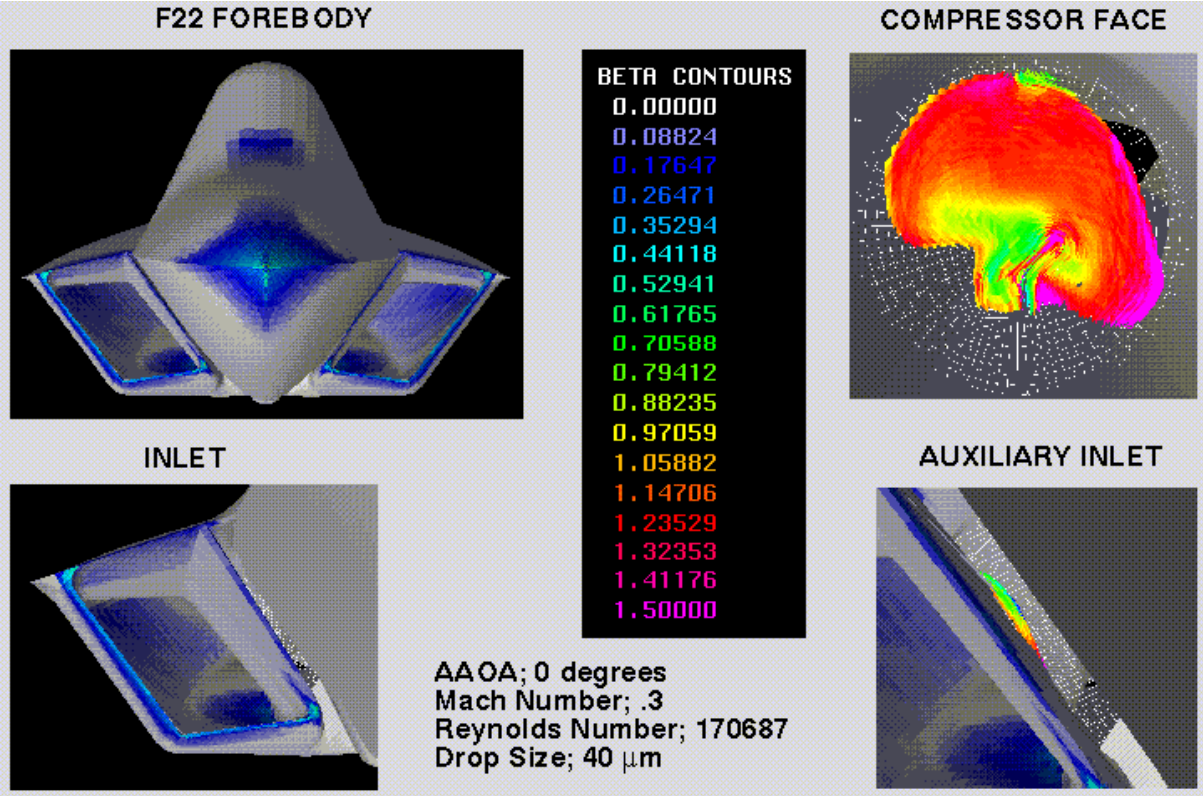
Icing Branch

5-29

The above depicts an ice accretion calculation for a multi-element wing at a typical scaled Reynolds Number for the Icing Research Tunnel under a fairly severe icing condition. The INS2D flow solver was used to generate the incompressible Navier-Stokes flow solution on the four block Chimera grid. Because of the relatively small size of the wing the severe icing condition results in large amount of ice. This figure and the previous one summarize the large effect scaling can have on ice shape.

Collection Efficiency Calculations For F22 Inlet



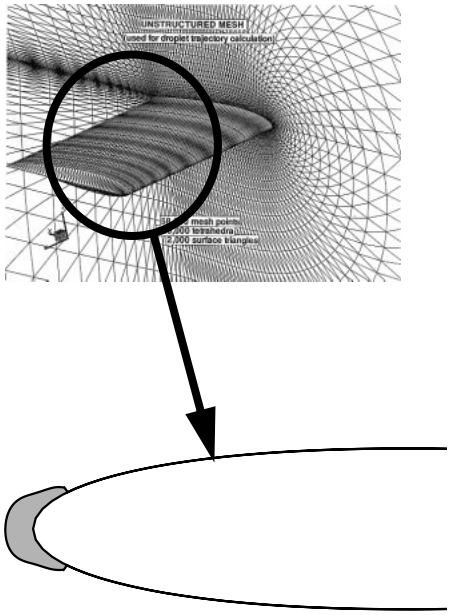
This figure shows the overall collection efficiency for an F22 inlet. The flow solution used was Euler based with 52 grid blocks and over a million points. The collection efficiencies were calculated using the scan-line Monte-Carlo method which required about 700,000 trajectories. These calculations required about a week on an SGI workstation. Collection efficiencies were generated at the nose, inlet lip, duct and compressor face in a single pass.

LEWICE3D Unstructured Grid Interface

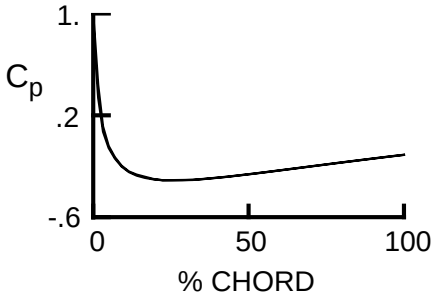
- This interface allows LEWICE3D to be used with any unstructured, multi-block, body-fitted Euler or Navier-Stokes flow solver.
- User inputs grid, flow and surface information in three separate ASCII files.
- Interfaces for Mavripolis and Rampant Unstructured grid codes have been developed.

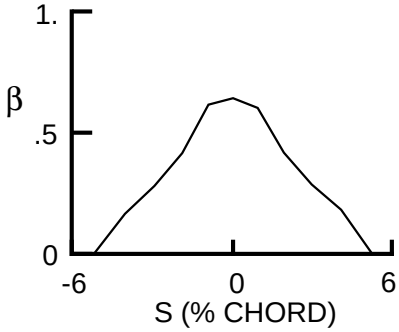
The unstructured grid interface allows the LEWICE3D code to handle multi-block, Euler or Navier-Stokes based flow solutions. A filter is provided with LEWICE3D that converts TECPLOT solution files (a common post processor supported by many unstructured flow solvers) into LEWICE3D files.

LEWICE3D Ice Accretion Calculation For A NACA-0012 Wing



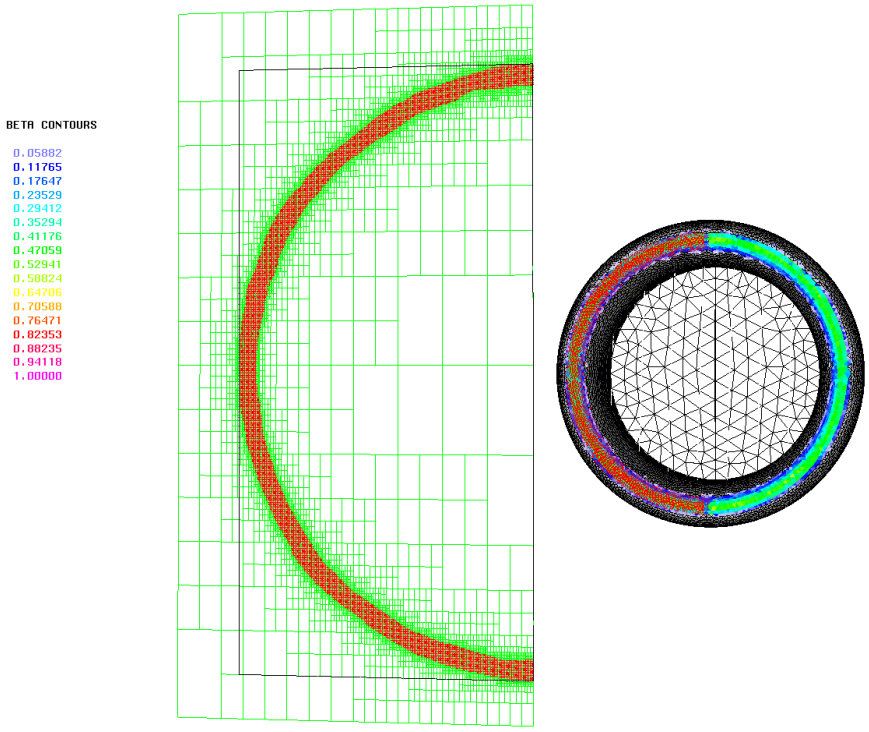
ICING CONDITIONS:
- Chord; .4406 m
- Airspeed; 63 m/s
- MVD; 20 µm
- LWC; .50 g/m^3
- Icing time; 10 min.
- Temperature; -18. C
- Pressure; 89867 Pa





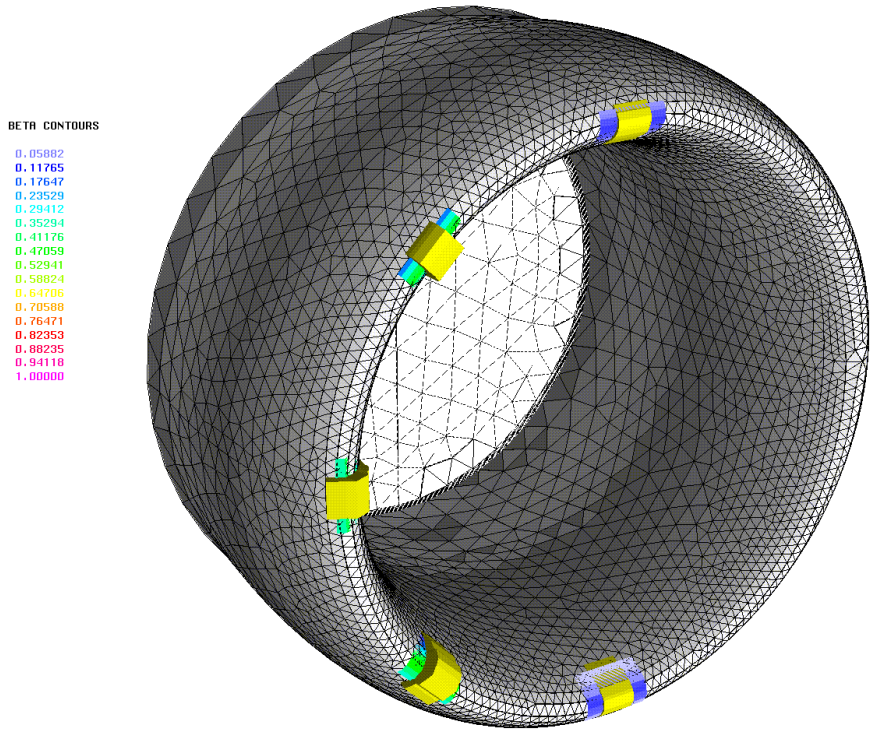
The above depicts an ice accretion calculation for a NACA-0012 wing tip. The flow solution was an Euler based solution generated using the Mavripolis unstructured flow solver. The ice shape is conformal and is indictative of a rime shape.

Collection Efficiency Calculations For Fluent Inlet



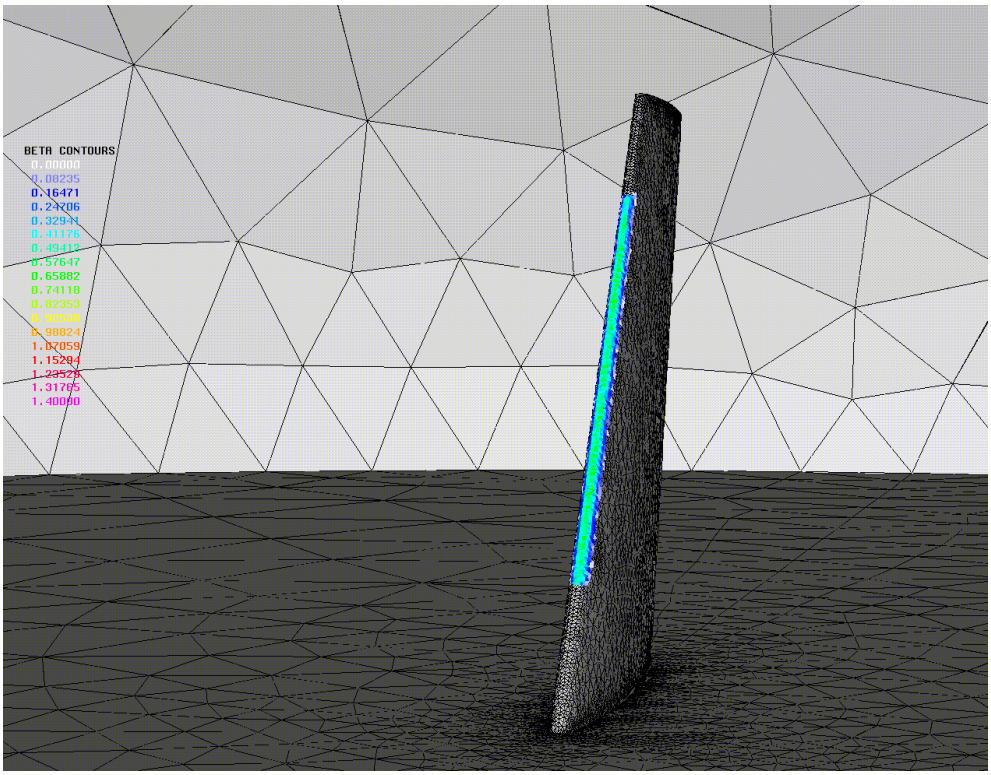
The above depicts the overall collection efficiency for an axi-symmetric inlet along with the trajectory release grid used in the unstructured collection efficiency calculation. The flow solution was generated using the Fluent unstructured flow solver. The red regions are the release points and landing points of the impacting trajectories. The symmetry of the inlet and flow condition result in an impingment region that is round and symmetric.

Ice Accretion Calculations For Fluent Inlet



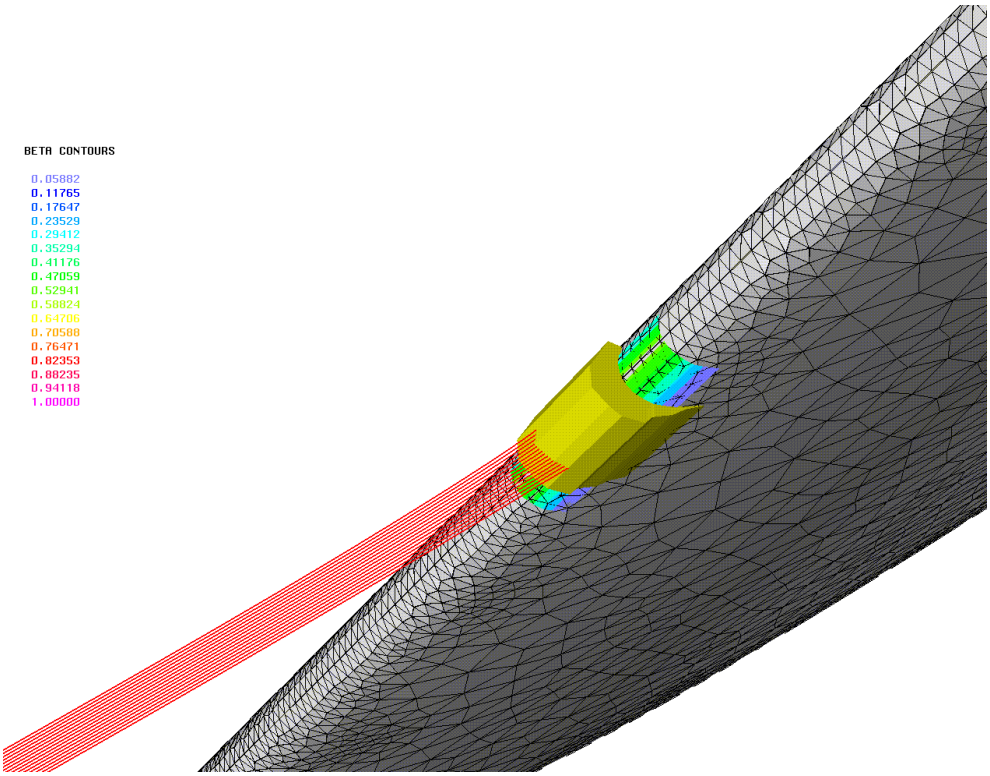
This figure shows the calculated ice shapes at several radial locations around the inlet. The flow solution was generated using the Fluent unstructured flow solver. The ice shape at every station except the upper and lower lips are similar due to the symmetry of the case. The reduced size of the ice shape at the upper and lower lip is due to a problem with the flow solution in these regions.

Collection Efficiency Calculations For Fluent Wing



The above depicts the overall collection efficiciency for a swept, tapered wing tip. The flow solution was generated using the Fluent unstructured flow solver. From the figure the decrease in collection efficiency as the root is approached due to the increase in chord length can be seen.

Ice Accretion Calculations For Fluent Wing



The above shows a series of impact droplet trajectories and the resulting ice shape for a swept, tapered wing tip. The flow solution was generated using the Fluent unstructured flow solver. The ice shape and the collection efficiency contours are very coarse due to the coarseness of the cells in the leading edge region.

Post Processors

- PLTLEW3D is a cross platform interactive post processor which allows interactive analysis of LEWICE3D icing parameters. This 2D post processor can generate postscript, ascii output for printing or for input into spreadsheets.

- LEWPLT3D is an interactive 3D post processor for viewing the 3D icing results on an SGI platform. The code allows the user to look at the 3D surface model, ice shapes, trajectories, streamlines, and grids.

- GRAHPCNV is a 3D post processor which converts LEWICE3D icing data into TECPLOT format. TECPLOT is available for any platform and can be used to plot the 3D surfaces, grids, ice shapes, trajectories and streamlines.

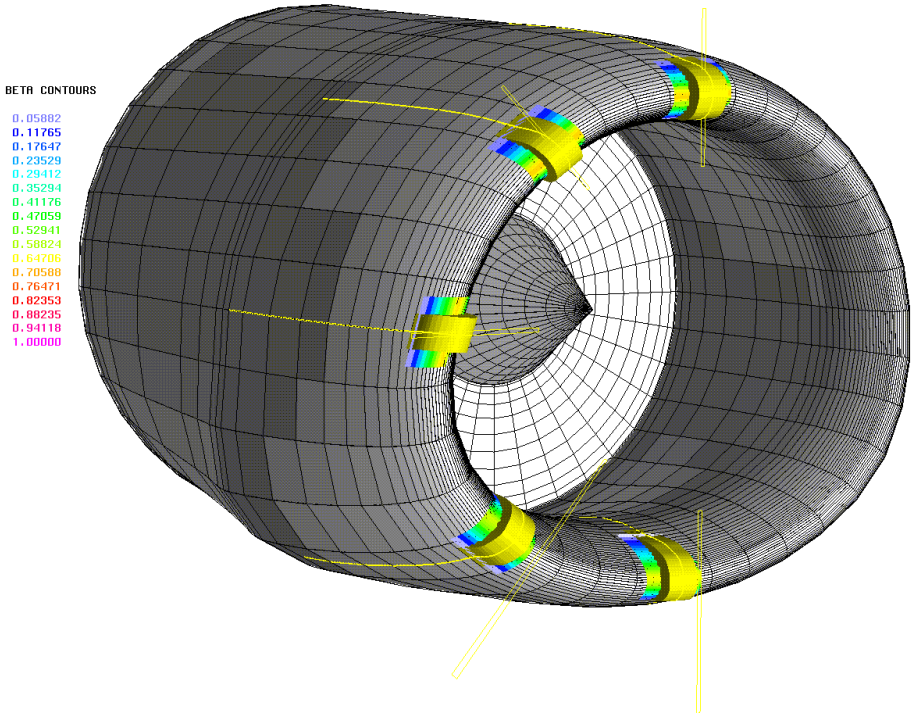
Three post processors are shipped with the LEWICE3D distribution for analyzing the LEWICE3D output. PLTLEW3D is a 2D post processor which allows viewing of the ice accretion data along the streamlines such as collection efficiency, heat transfer coefficient, etc. PLTLEW3D is portable to any platform and can be used to dump postscript files for printing or ASCII files for use in spread sheets. LEWPLT3D is an SGI based interactive post processor for veiwing the 3D icing results. The post processor can be used to view trajectories, ice shapes, streamlines, collection efficieincy contours, ICEGRID grids, surface models and grids. GRAPHCNV is a batch post processor which converts LEWICE3D output into TECPLOT format for viewing in that graphics package. TECPLOT can be used to view trajectories, ice shapes, streamlines, collection efficieincy contours, ICEGRID grids, surface models and grids. TECPLOT (available from AMTEC) is available for any platform.

Demonstration

- LEWPLT3D
- PLTLEW3D

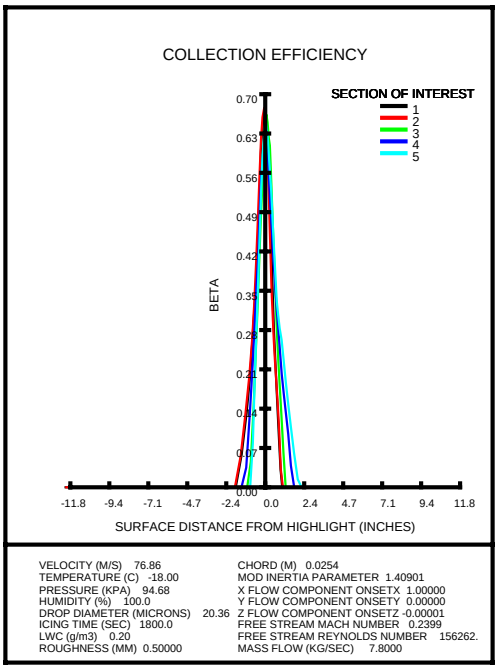
The demonstration will involve running the LEWICE3D code for the Boeing 737-300 inlet case which was presented earlier. The LEWICE3D case will be executed and the post processors LEWPLT3D and PLTLEW3D will be used to view the results.

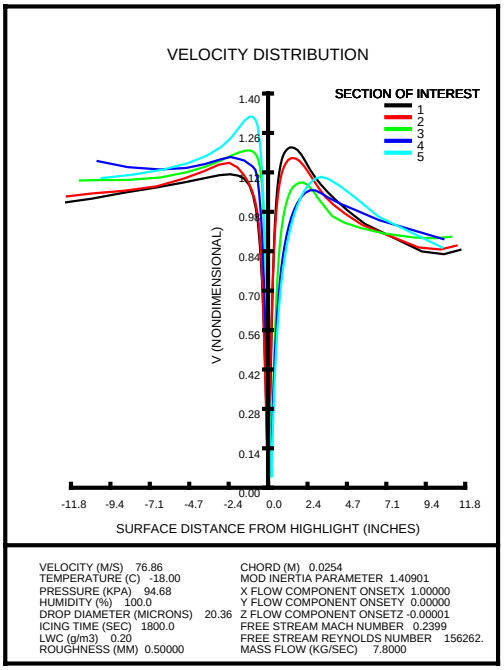
LEWPLT3D OUTPUT FOR 737-300 INLET



The above is a typical output for the LEWPLT3D interactive post processor. Plotted along with the VSAERO panel model are streamlines, ice shapes and collection efficiency contours for each of the five calculated ice shapes. Also included in the image are the yellow boxes describing the sections-of-interest used in the collection efficiency calculation.

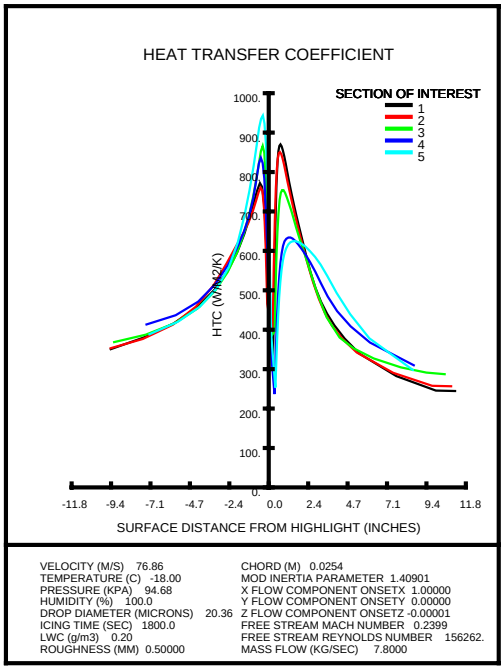
PLTLEW3D OUTPUT FOR 737-300 INLET

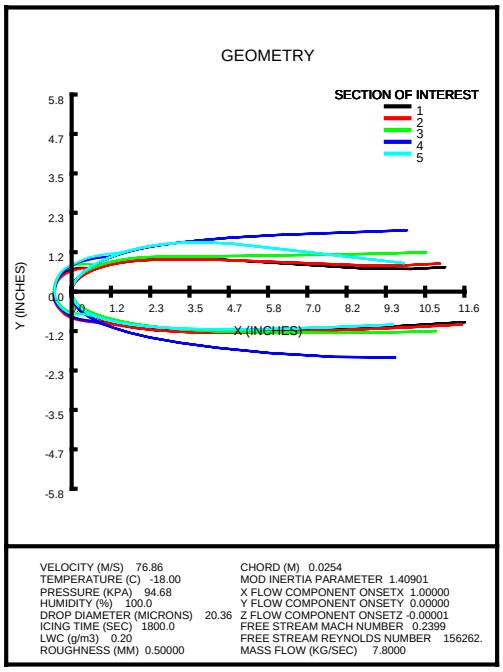




Examples of the color postscript output from the PLTLEW3D post processor are given above. The color plots depict the collection efficiency and edge velocities distributions for each of the five sections-of-interest.

PLTLEW3D OUTPUT FOR 737-300 INLET





Examples of the color postscript output from the PLTLEW3D post processor are given above. The color plots depict the heat transfer coefficient distributions and the ice shapes for each of the five sections-of-interest.

Future Plans

- ICE3D

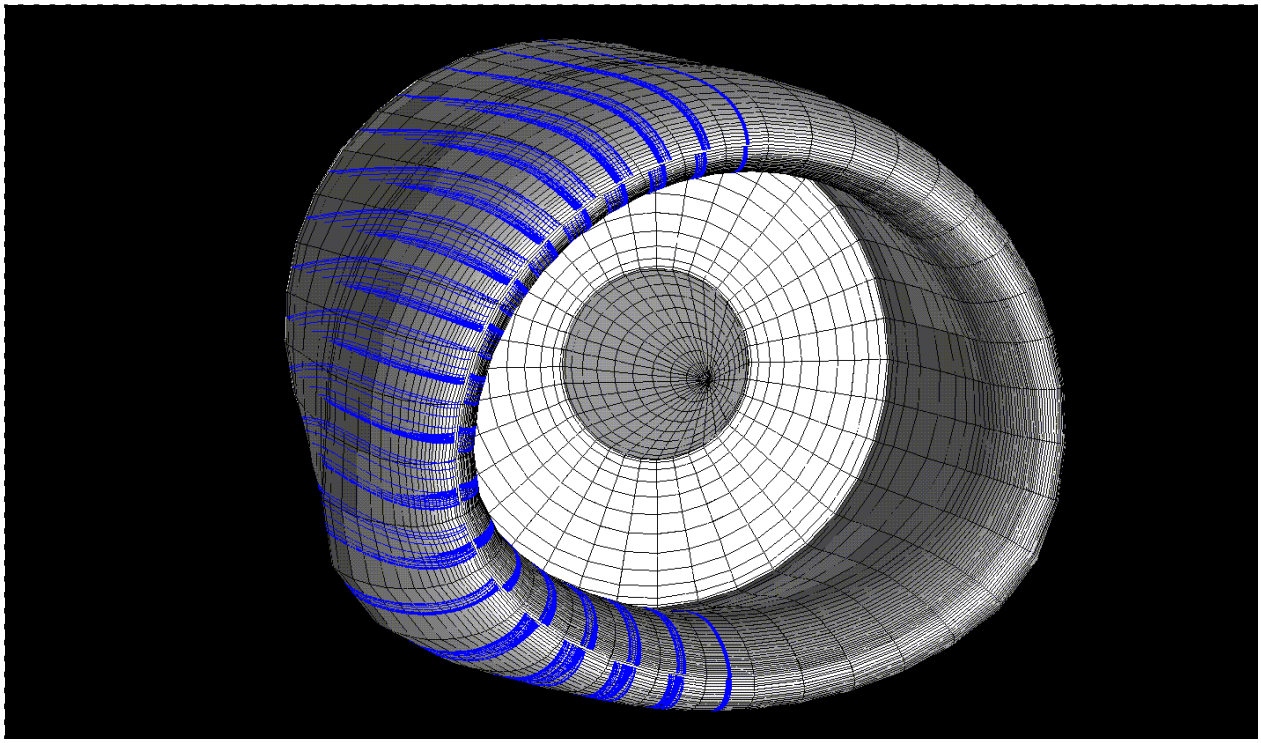
A discussion of the 3D ice accretion code ICE3D will be given along with some preliminary calculations. ICE3D applies the LEWICE3D strip approach to every panel on a surface to generate a new 3D surface. This is different from the LEWICE3D code which generates “strip” ice shapes at specified sections-of-interest. LEWICE3D has no provisions for modifying the input surface or grid file.

ICE3D Ice Accretion Code

- Code generates full 3D ice accretion for surfaces using LEWICE3D strip approach at every surface panel.
- Code requires surface model, surface velocities and collection efficiencies. Optionally surface Heat Transfer Coefficients may be input.
- Code can be interfaced with panel codes, and structured or unstructured Euler or Navier-Stokes flow codes.
- Generates new iced surface from old surface and ice thickness distribution generated from strip approach applied to each panel. This new surface can be used for performance evaluations or in the next iteration of a multi-step calculation.

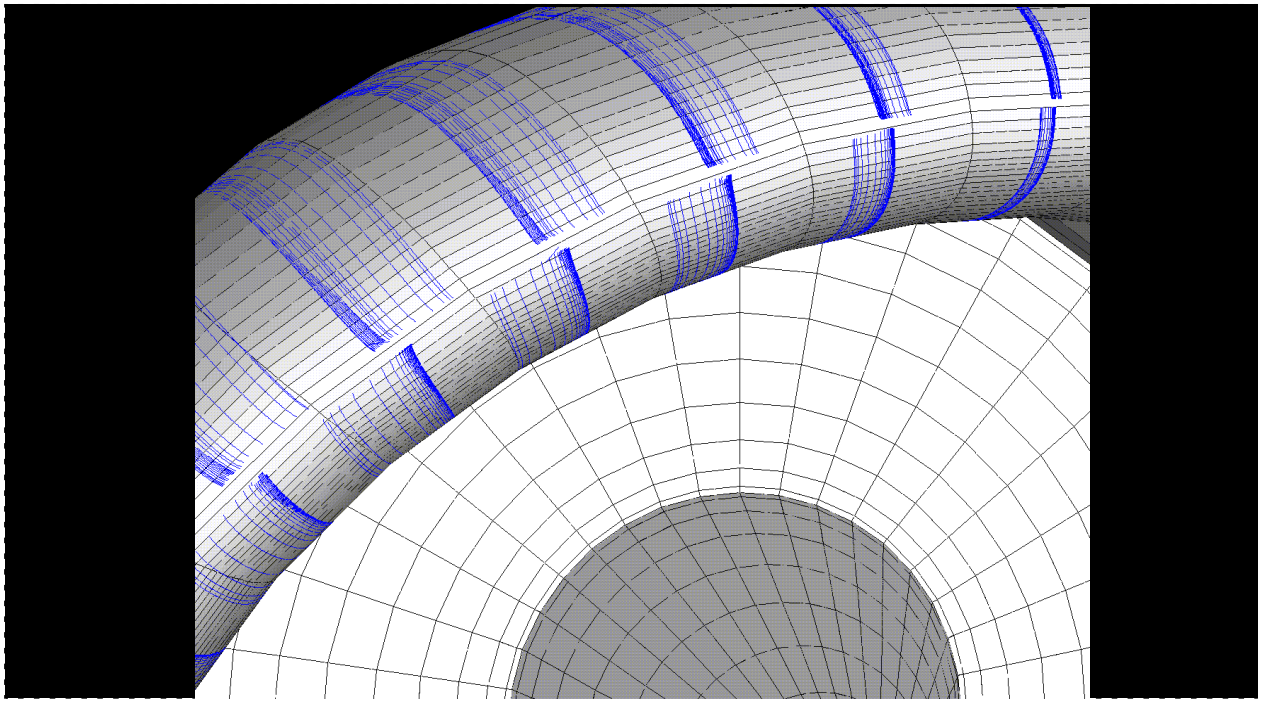
ICE3D applies the LEWICE3D strip approach to every panel on a surface to generate a new 3D surface. The program does not have a trajectory code and requires the input of collection efficiency data at every panel on the surface along with surface velocities and grids. The code calculates surface streamlines and uses input surface velocities and collection efficiencies to calculate surface ice thickness distributions using the LEWICE3D ice accretion module.

Surface Streamlines For A Boeing 737-300 Inlet



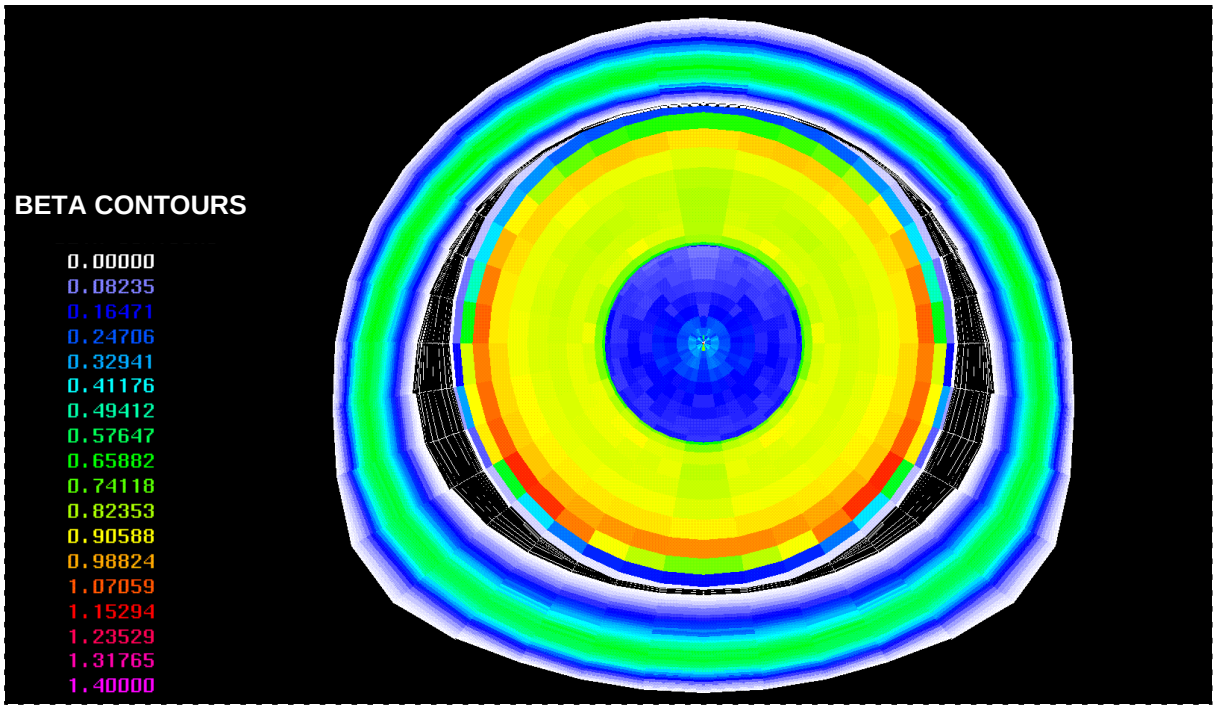
The above shows the surface streamlines generated using ICE3D on the Boeing 737-300 inlet. Note that there is one streamline for each panel on the surface.

Surface Streamlines For A Boeing 737-300 Inlet



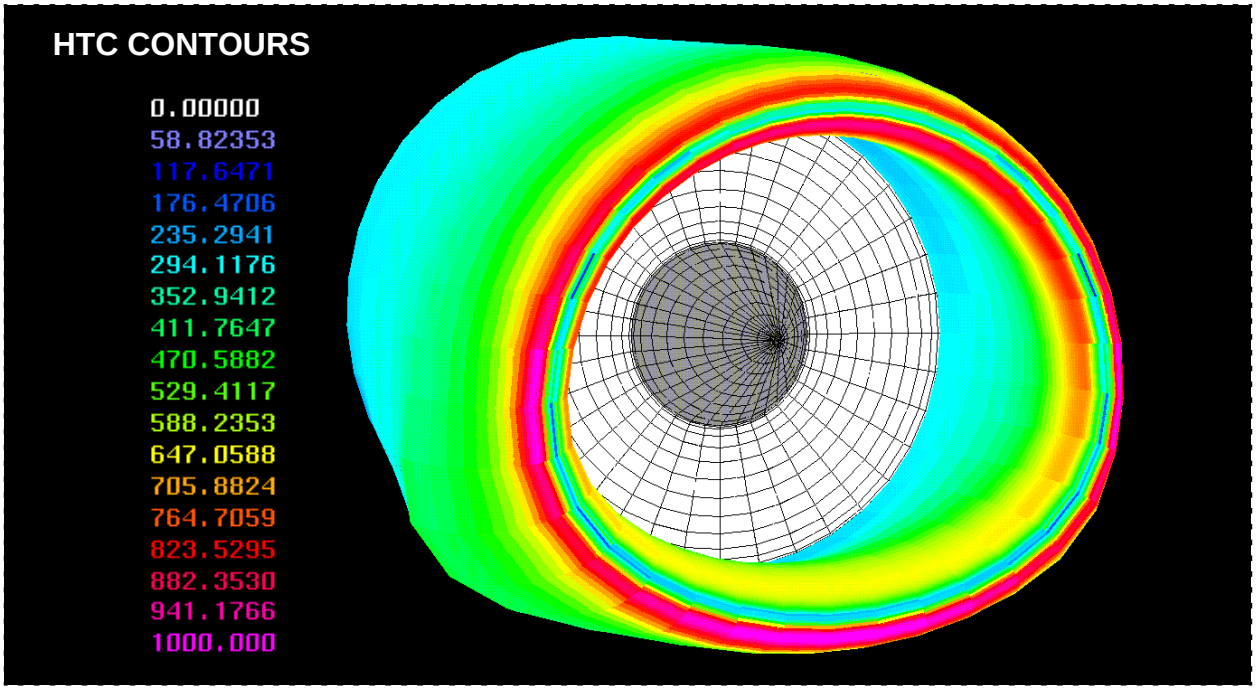
The above shows a close up of the surface streamlines generated using ICE3D on the Boeing 737-300 inlet. Notice the irregularity of the streamlines near the stagnation lines. Calculating streamlines accurately in the stagnation region accounts for most of the challenge in the ICE3D algorithm.

Collection Efficiency Calculations For A Boeing 737-300 Inlet



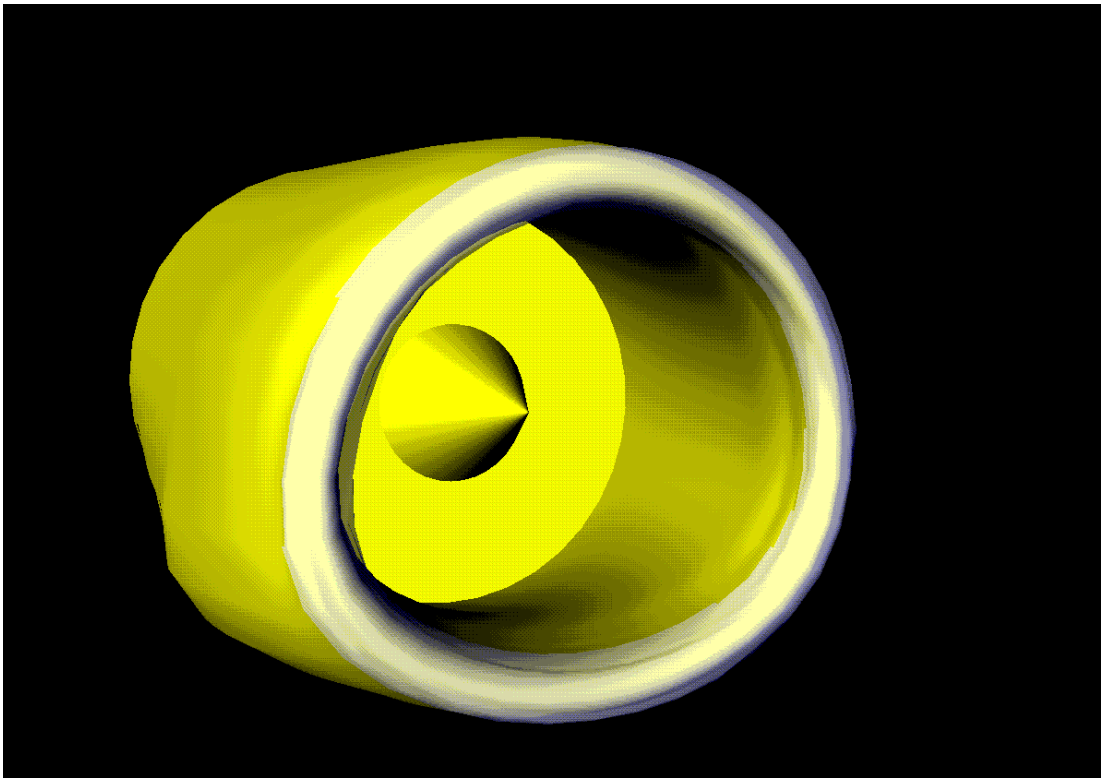
The above shows the overall collection efficiency for the Boeing 737-300 inlet. This collection efficiency map, which is required by ICE3D as input, was generated using the LEWICE3D code.

Heat Transfer Coefficient Calculations For A Boeing 737-300 Inlet



The above shows the heat transfer coefficient distribution for the Boeing 737-300 inlet. This distribution was generated by the ICE3D program. The heat transfer coefficient distribution can be optionally input into ICE3D.

Ice Accretion Calculations For A Boeing 737-300 Inlet



The above shows the 3D ice shape generated by the ICE3D program. Full 3D ice shapes are needed if multi-time stepping is to be attempted. From the full 3D ice shapes new surface models or grids can be generated for use in the next time step.

Conclusions

- The LEWICE3D 3D grid based ice accretion code has proven flexible in handling any type of flow solution and icing problem.
- The LEWICE3D code is fast and portable producing ice shapes in minutes on a modern PC.
- The graphics post processors included with the distribution allow quick and easy analysis of the complex and cumbersome data output from the LEWICE3D program.
- The ICE3D program represents the next step in developing a full 3D multi-step ice accretion program.

The LEWICE3D is flexible with the capability to handle any 3D flow solution and icing problem. The LEWICE3D code and it's associated post processors can been ported to any platform and execution times on a modern PC are typically less than 5 minutes. The associated post processors greatly simplify the analysis of the LEWICE3D output data. The ICE3D program uses the LEWICE3D strip approach at every surface point to generate a new 3D surface. This surface can be used for multi-step 3D ice accretion calculations. The incorporation of multi-stepping into the 3D ice accretion codes represents the next challenge in code development.